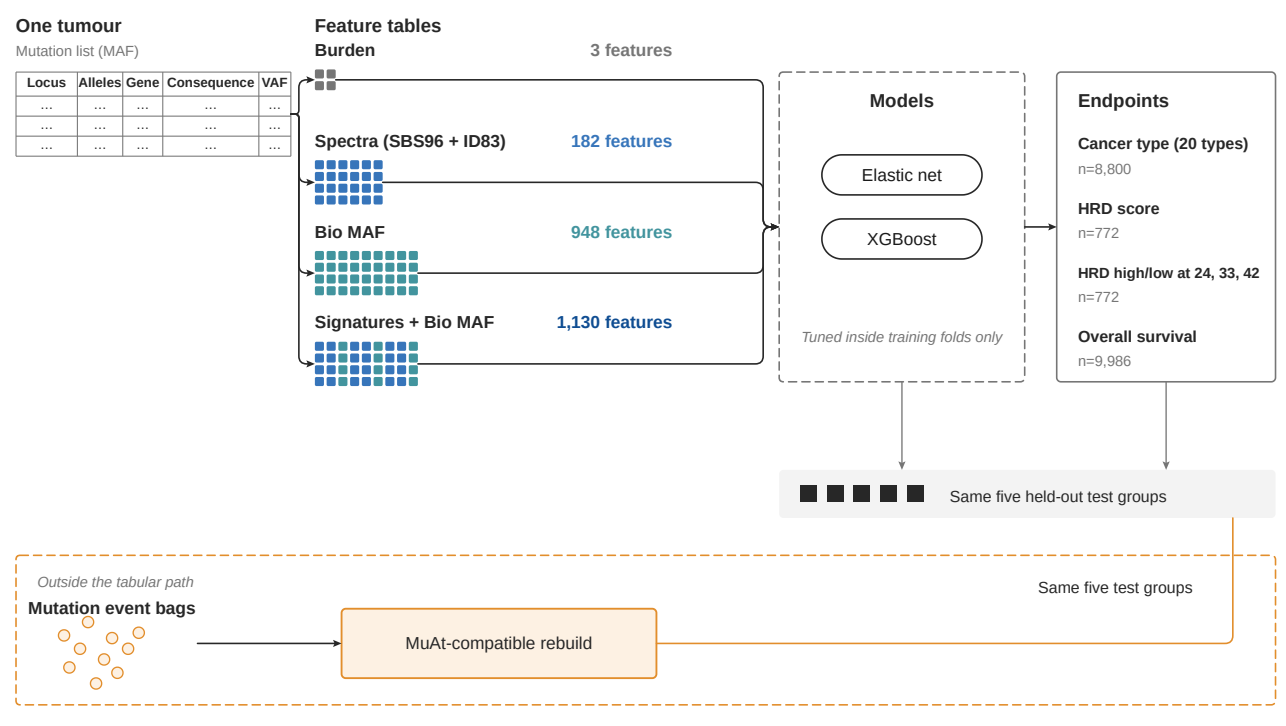

One tumour
Mutation list (MAF)
Locus
Alleles
Gene
Consequence
VAF
Feature tables
Burden
3 features
Spectra (SBS96 + ID83)
182 features
Bio MAF
948 features
Signatures + Bio MAF
1,130 features
Models
Elastic net
XGBoost
Tuned inside training folds only
Endpoints
Cancer type (20 types)
n=8,800
HRD score
n=772
HRD high/low at 24, 33, 42
n=772
Overall survival
n=9,986
Same five held-out test groups
Outside the tabular path
Mutation event bags
MuAt-compatible rebuild
Same five test groups